# Image synthesis from themes captured in Poems using latent diffusion models

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***Abstract*—Due to the presence of complex literary devices such as metaphors and imagery, poetry can be difficult to comprehend. This is especially true for literary works of classical poets like Kaalidasa that employ intricate, often conflicting themes which tend to be particularly tedious to interpret and decipher. The beauty of these works of art tends to get lost in translation. A visual representation in the form of images corresponding to the various themes in the poetry, greatly aids in providing a clearer understanding of the meaning and imagery described.**

**The main aim here is to make classical poetry more accessible by generating detailed images that capture and depict the metaphors and themes used in various works of literature. The core task in this paper is to employ novel machine learning (NLP) techniques to detect and extract the central themes and keywords from the poems that encapsulate the essence of the literary work. This is done using transformer models fine-tuned specifically on a summarization dataset, that generate an abstractive summary of the segment of input text. Maintaining context while doing so is essential to the accuracy of the images being generated. Further, this summary is then provided as an input to a Latent Diffusion Model to generate detailed images corresponding to the poetry.**

**The goal of the project is to make it easier to consume and enjoy classical works of literature by providing additional context and information in the form of images complementing the poetry.**

***Index Terms*—Latent Diffusion Models, semantic structure, Transformers, diffusion processes, Text Simplification, Abstractive Summarisation, Human Evaluation, Computed Relevance, Visual Relevance, accompanying images, comprehension, optically satisfying, relevance, significance, poetry, imagery, semantics, memory.**

## I. INTRODUCTION

It is scientifically proven that images and videos are more effective at conveying what plain text is aiming to express. Visual representations aid in better comprehension, make communication quicker and simpler, and act as a stimulator for emotions. The aforementioned factors help retain information for a longer duration of time and create a stronger impact on the reader/viewer.

Because of a linguistic barrier, ancient literature and poetry in languages such as Hindi, Kannada, and Sanskrit are not receiving the attention they deserve from current audiences. The beautiful complexities of great poets' writings, such as Kabir, Tulsidas, and Kalidasa, can be scary to modern readers. Kalidasa in particular uses an abundance of metaphors and imagery in his work that requires the reader to closely decipher and carefully interpret his poetry. This provides a great opportunity to employ another medium to aid in understanding his literary work.

Furthermore, poetry as an art form is often challenging to interpret due to complex literary devices like metaphors. A visual representation of the various themes in poems greatly aids in providing a clearer understanding of the meaning and imagery described. Our project aims to generate detailed images representing the themes used in popular poems and literary work.

This paper is targeted at making ancient and classical literary works more accessible and easier to decipher with the support of descriptive images that accurately represent the imagery used in the works. The core objective is to efficiently summarize the text while preserving the underlying context anf further generate detailed image depictions of the poems.

Exhaustive literature survey to explore and gauge various methods to accomplish our goal has been conducted. Our approach includes two major components – the language processing model and the image generation model. For the language processing component, abstractive summarization was implemented using a pretrained transformer model. Based on our literature survey, this technique appears to be the most recent and effective in preserving context while simplifying and condensing the text into comprehensive summaries. Next, for the image generation component, a Latent Diffusion Model has been implemented. In contrast to GANs (which was the previously proposed method for the image synthesis task), Diffusion Models built from a hierarchy of denoising autoencoders prove to be effective in modelling complex, even multimodal distributions. The images generated using this model are highly detailed images based on the summary obtained from the language processing component.

## II. RELATED WORKS

Image synthesis from textual inputs has been an area of active research in recent years. The following papers have

made significant contributions to this field:

Schulder and Hovy [1] proposed a method for metaphor detection through term relevance using textual data. Dinh and Gurevych [2] introduced a neural network-based approach for token-level metaphor detection. Both of these works are relevant for our study as they provide a means to identify themes in poems that could be used for image synthesis.

The Microsoft Common Objects in Context (COCO) dataset [3] is a popular benchmark for image recognition and synthesis tasks. It contains over 330,000 images with object annotations and captions. This dataset has been used in various image synthesis studies, including Das and Saha's work on self-supervised image-to-text and text-to-image synthesis [4]. Dosovitskiy et al. [5] presented a method for generating images of objects using convolutional networks. Min et al. [6] proposed a deep learning-based approach for short story generation from images. Both of these works demonstrate the feasibility of using neural networks for image synthesis.

Angrosh et al. [7] introduced a conditional random field-based method for identifying context in related work sections of research papers. This method is relevant for our study as it could be used to automatically identify relevant themes in poems.

Li et al. [8] proposed a controllable text-to-image generation approach that allows the user to specify visual attributes of the generated image. This work is relevant for our study as it provides a means to control the style and content of the generated images.

Frolov et al. [9] presented a comprehensive review of adversarial text-to-image synthesis methods. Donahue et al. [10] proposed a long-term recurrent convolutional network for visual recognition and description. Cheng et al. [11] introduced an approach for verbal-guided image parsing. Mansimov et al. [12] proposed an attention-based method for generating images from captions. Nam et al. [13] introduced text-adaptive generative adversarial networks. Zhang et al. [14] presented the StackGAN approach for text-to-photo-realistic image synthesis. All of these works are relevant for our study as they provide different approaches for text-to-image synthesis.

Liu et al. [15, 18, 19] proposed a generative adversarial network-based approach for abstractive text summarization. Beladev et al. [16] presented a language-agnostic model for transforming text into sentences. Balabantaray et al. [17] introduced a method for text summarization using term weights. These works are relevant for our study as they provide approaches for generating textual summaries of the poems that could be used for image synthesis.

## III. DATASET

This chapter talks about the dataset which is going to be used in the project. The dataset used in the image generation module is caption to image. So each image has a caption to it and the model is trained on this dataset. English translations of Kalidasa's poem have been mainly used to test the framework.

3.1 Overview There are a lot of other text-image datasets but LAION-400M is one of the most robust ones among them.

LAION-400M(open source text-image dataset): The dataset is built using OpenAI's Clip's cosine similarity between the text and image and the text to image pair whose cosine similarity is above 0.3 was only considered rest all the pairs were discarded. 0.3 was determined from trial and error basis. This was done to remove not so appropriate text-image pairs and to improve the overall semantic image-text content matching. This dataset has 413 million text-image pairs and has been used to train the image synthesis component.

Poems: Since the main aim is at generating images from poems which can describe what is that the poet is trying to make the viewer visualise, so that is why to test the framework kalidasa's work has been used since it has a lot of imagery described in the poem from which aesthetically pleasing images can be formed. The english translations of the poems are openly available on the internet.

The dataset used for the project is discussed, specifically for the image generation module, which is based on a caption-to-image dataset. Each image in the dataset is associated with a corresponding caption, and the model is trained on this dataset. To test the framework, English translations of Kalidasa's poem have been used.

There are various text-image datasets available, but LAION-400M is considered one of the most robust ones. This open-source text-image dataset was built using OpenAI's Clip's cosine similarity between the text and image. Only those text-to-image pairs whose cosine similarity was above 0.3 were considered, and the rest were discarded. The value of 0.3 was determined from a trial and error basis. This was done to remove not-so-appropriate text-image pairs and improve the overall semantic image-text content matching. The dataset comprises 413 million text-image pairs and has been used to train the image synthesis component.

Since the paper aims to generate images from poems that describe what the poet wants the viewer to visualize, English translations of Kalidasa's poems were used to test the framework. Kalidasa's work is known for its rich imagery, making it ideal for generating aesthetically pleasing images. These translations are readily available on the internet.

## IV. PROBLEM STATEMENT AND INITIAL APPROACH

Research has been conducted in the fields of image synthesis and text summarization but research involving text processing of classical works of literature is limited. Due to the presence of complex literary devices such as metaphors and imagery, classical poetry can be difficult to comprehend. The beauty of these works of art often gets lost in translation. A visual representation in the form of images corresponding to the various themes in the poetry, greatly aids in providing a clearer understanding of the meaning and imagery described.

The aim is to generate an image or a set of images from stanzas or verses of poems. The English translations of works of Kalidasa were primarily dealt with as his literary style prominently uses metaphors and comparisons which can be enjoyed and understood more easily when visualized.

The model proposed contains two parts: an abstractive summariser using a transformer model for generating captions in the form of summaries and a Latent Diffusion Model for image synthesis. The high level workflow consists of passing a stanza or stanzas of a poem into the caption generator which then produces an abstractive summary describing the scenes for the image synthesizer to generate detailed images.

The caption generator utilizes abstractive summarization as a method to obtain the descriptive aspects and central themes of the text and convert them into summaries that serve as captions. A Transformer model has been employed to achieve this. The Transformer uses a self-attention mechanism where attention weights are calculated using all the words in the input sequence at once, hence facilitating parallelization.Two summarization models from the transformers library were considered: BART trained on the CNN/Daily Mail News Dataset and T5 trained on the c4 Common Crawl web corpus.

The image synthesis is achieved using a Latent Diffusion Model. The abstractive summary would be the input to this model which will output highly detailed descriptive images. Since Diffusion Models consume hundreds of GPU days and computational resources are limited, here, Diffusion Models have been applied in the latent space of pretrained autoencoders. The use of LDMs is justified because they are capable of preserving the semantic structure of the data while generating highly detailed and optically pleasing images.

Compared to earlier image generation techniques, LDMs use diffusion processes in latent space instead of pixel space, resulting in a more dependable and memory-efficient approach. This means that the generated images are more accurate and require less computational resources to produce.

Moreover, LDMs incorporate semantic data from Transformers, which is a highly advanced language model, making them capable of understanding the context of the text and generating images that are contextually appropriate. This ensures that the images are not only visually appealing but also relevant to the text they are generated from.

In the case of the example text provided, "The summer's clear late nights are the perfect time to enjoy a rooftop cocktail," LDMs can capture the essence of the text and generate an image that accurately represents the scene described in the text, such as a person enjoying a cocktail on a rooftop on a summer night.

Therefore, the use of LDMs for image generation from text is justified because they are highly efficient, contextually aware, and can generate optically pleasing images while preserving the semantic structure of the data.

## V. PROPOSED SOLUTION

***5.1 Overview:*** At an abstraction level the whole methodology is implemented such that when a poem is given as an input, the output should be generated such that it is able to visually describe the scenery the poet is trying to make the reader visualise through the poem. There are various modules which have been implemented and the details of all these will be covered in the next subsection of this chapter.

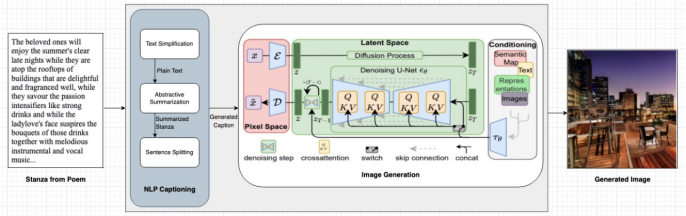


Fig. 1. Overview of the pipeline being implemented.

***5.2 Modules:***

*1)* ***5.2.1 Text simplification:*** Text simplification is a way of language processing that aims to make written text simpler to read and understand. This can be helpful for people with limited reading skills, such as those learning a second language or those with dyslexia. Text simplification methods can include techniques like rephrasing complicated sentences, reducing the use of rare words, and replacing longer, more complex words with shorter, simpler ones. Some text simplification methods are fully automated, while others need human editors to make sure the simplified text is still grammatically correct and maintains the original meaning of the text.

There is no need for labelled simplification data with the multilingual sentence-simplification system known as MUSS. It employs a novel methodology that develops robust models using data on sentence-level paraphrase rather than data on simplification. These models modify properties like length and lexical difficulty at inference time using unsupervised pre training and programmable generation techniques. Additionally, MUSS offers an approach that does not require labelled data and uses semantic sentence embeddings to mine paraphrase data from Common Crawl in any language. As a result, MUSS can shorten phrases in any language without the need for labelled training data.

*2)* ***5.2.2 Abstractive summarization:*** In order to obtain context from the stanzas of a poem summarization is used. As poems in general use a significant amount of abstract concepts to convey their messages, abstractive summarization was chosen over extractive to generate the required caption.

Two summarization models from the transformers library were considered: BART trained on the CNN/Daily Mail News Dataset and T5 trained on the c4 Common Crawl web corpus. Both use Transformer-based neural machine translation architecture to generate the summary from a given text and after testing both models on the data from Kalidasa's works, T5 provided more abstract summaries less reliant on the specific words given in the aforementioned text.

When configuring the parameters in the text summarization process summary length was an important factor. Captions obtained have been specified to 40 percent the length of the given stanza to ensure the description is not unnecessarily verbose. Additionally the OS environment is set to utilise the GPU to improve performance. An accurate description from a simplified stanza of a poem is thus obtained.

*3)* ***5.2.3 Paragraph splitting****:* As poems do not necessarily follow the grammatical rules set out to define the beginning and end of a sentence, an alternate mechanism must be employed in order to split a given stanza into its respective lines. The model takes two lists of words as input and decides whether to conjoin the line using positive and negative cases as described below. It is trained on the wikisent2 dataset containing sentences from the various articles on the wikipedia website.

A positive pair is defined as a pair of sentences that should be split and remain two different sentences with '.' between them and a negative pair is a pair of sentences that should be merged and considered as one sentence, with no '.' between them. Positive pairs are created using the 'nltk.sent tokenize function', splitting the text by '.', which is true for most of the cases and will let the model learn the mechanism and role of the '.' character while negative pairs are created by splitting sentences in the middle, and the min sentence length for splitting parameter is used to define the minimum length of sentence that would be split for the negative pairs.

For the first sentence a forward LSTM layer is applied and for the second sentence a backward LSTM layer. This is done in order to predict whether the line should have a '.' in the middle, so it is expected for the forward pass of the first sentence to embed a pattern of an end of the first sentence and the backward pass of the second sentence to embed a pattern of its beginning. Then both embeddings are concatenated to represent the ending and beginning meaning of the two sentences in order to predict if they should be split.

*4)* ***5.2.4 Image synthesis****:* As poems describe a lot of imagery,creativity and scenic description of a lot of things which have been taken in the context of the poem, therefore a robust model which could give justice was required.The latest and state of the art deep learning model which uses diffusion modules as a fundamental basis was used. This latent diffusion model is trained on the LAION-400M dataset(The details of this has been included in the previous sections).

In recent times GAN's have pioneered the image synthesis domain but recent findings have shown that diffusion models outperform GAN's since diffusion models don't suffer with phenomenons like mode collapse, lack of diversity in image generation and high training time, therefore diffusion models tend to perform better and outperform even the state of the art GAN's. The two primary process domains in a classic

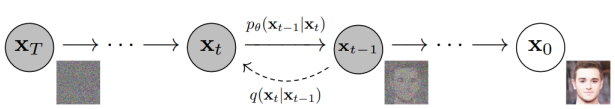


Fig. 2. Progression of diffusion models (Ref. [18])

diffusion model are forward diffusion and reverse diffusion. The quantity of noise in the image gradually becomes worse throughout the Forward Diffusion stage until it is completely random noise. The alternate technique recovers the data from the Gaussian noise by gradually reducing the anticipated noise at each time step using a succession of Markov Chains.

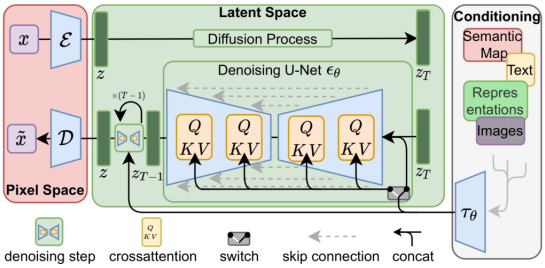


Fig. 3. LDM (Ref. [18])

A recently proposed approach combining the perceptual power of GANs, the detail retention of Diffusion Models, and the perceptual power of Diffusion Models with the semantic power of Transformers. This approach is known by the authors as "Latent Diffusion Models" (LDM). Compared to the earlier varieties, LDMs have differentiated themselves as being more dependable and efficient. They offer a variety of very detailed graphics that preserve the semantic structure of the data, as well as using less memory than prior methods. In essence, an LDM incorporates semantic data from the Transformers and runs diffusion processes in latent space rather than pixel space.

## VI. RESULTS AND INFERENCES

TABLE I

| | | |
|---|---|---|
| The beloved ones will enjoy the summer's clear late nights while they are atop the rooftops of buildings that are delightful and fragranced well, while they savour the passion intensifiers like strong drinks and while the ladylove's face suspires the bouquets of those drinks together with melodious instrumental and vocal music... | The summer's clear late nights are the perfect time to enjoy a rooftop cocktail. | |

Above is an example of an image generated based on one of the stanzas of Kaalidaasa's work (as shown in the first column). The poem contains rich imagery including mentions of rooftops and late nights. However, the poem in itself is difficult to interpret due to convoluted comparisons and non-linear sentences. The second column contains the text after the poem has been passed through the Text Simplification and Abstractive Summarisation modules. Here, a caption is generated that encapsulates the essence of the poem. This is one of the captions generated for the given stanza. Further, the third column shows the image that has been generated from the caption.

Below are some more examples of the images generated for various stanzas. The images are optically pleasing while capturing the meaning and imagery described in the poems.

The technique used to measure the accuracy and appropriateness of the images is Human Evaluation. On extensive experimentation, it was found that other methods like object detection followed by keyword matching in the captions or reversing the process by converting the image into captions and then comparing that to our captions proved to be ineffective because both these methods fail to recognize the context of the poem thereby misinterpreting the appropriateness of the image. Human Evaluation appears to be the best technique as the images are intended for the reader and viewer to consume. Within Human Evaluation, two metrics have been used. The first one is Computed Relevance, which is the ratio between the number of objects, attributes or descriptive keywords present in the generated caption and the number of these present in the generated images. This is the quantitative metric. The second metric is Visual Relevance which is a score on a scale of 1 to 5 that qualitatively assesses how appropriate or relevant the generated image is to the poem it is supposed to describe.

TABLE II

| | | |
|---|---|---|
| Where horses, as dark as leaves, rival the steeds of the sun; where elephants, as tall as mountains, pour forth showers, like you, from the pores of their temples; and where the foremost warriors stood in battle against the ten-faced one, the splendour of their ornaments surpassed by the scars of the wounds from Candrahasa; | where the scars of Candrahasa's battle with the ten-faced one can be seen on the faces of the warriors. | |

TABLE III

| | | |
|---|---|---|
| Where horses, as dark as leaves, rival the steeds of the sun; where elephants, as tall as mountains, pour forth showers, like you, from the pores of their temples; and where the foremost warriors stood in battle against the ten-faced one, the splendour of their ornaments surpassed by the scars of the wounds from Candrahasa; | elephants are tall as mountains | |

Based on the below data, average Computed Relevance is 75 percent and average Visual Relevance is 69 percent.

## VII. CONCLUSION

Poems prove to be difficult to interpret as it has been found during attempts to process them. As a result it is believed that generating accompanying images can vastly improve comprehension. Using text simplification and Abstractive Summarization, captions are obtained from the text and the Latent Diffusion Model synthesizes the corresponding images. Relevance of the given images is evaluated based on the processed text and results show that they are of favorable quality. Thus can be safely concluded that the images being generated by the developed pipeline is optically satisfying with metrics to qualitatively measure their relevance and significance.

TABLE IV
ACCURACY

| Line Number | Computed Relevance | Visual Relevance |
|---|---|---|
| 1 | 0.60 | 0.67 |
| 2 | 0.83 | 0.76 |
| 3 | 0.77 | 0.56 |
| 4 | 0.68 | 0.60 |
| 5 | 0.68 | 0.79 |
| 6 | 0.82 | 0.74 |
| 7 | 0.53 | 0.48 |
| 8 | 0.82 | 0.70 |
| 9 | 0.52 | 0.51 |
| 10 | 0.88 | 0.78 |
| 11 | 0.96 | 0.76 |
| 12 | 0.58 | 0.57 |
| 13 | 0.50 | 0.65 |
| 14 | 0.88 | 0.73 |
| 15 | 0.57 | 0.52 |
| 16 | 1.00 | 0.97 |
| 17 | 1.00 | 0.94 |
| 18 | 0.83 | 0.77 |

TABLE V
FINAL AVERAGE RESULTS

| Computed Relevance | Visual Relevance |
|---|---|
| 75% | 69% |

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